Ethical Side of AI Being Used in Finance

Artificial Intelligence (AI) is an essential tool that is being used worldwide. It becomes more ubiquitous everyday with new features. The most used type of AI is Large Language Models (LLM) in daily life. However AI has more capabilities than just chatting, such as making deep analysis depending only on numbers. For example there are AI bots that are making financial analysis'. These bots have a lot of potential for making highly accurate forecasts. This potential may result in the removal of the human factor from the financial markets. This removal would also result in the lack of emotions in the market and reduce finance to mere mathematics by charming away all its depth and dimensions.

To be able to learn more and improve my programming skills I have created a stocks forecasting AI bot. Throughout this process I had made a lot of research and experiments to find the best parameters for my bot. I tried several different scenarios changing every parameter and trying new things so that it does not overfit nor underfit. I used Long-Short Term Memory (LSTM) which is a model that is being used for the analysis of numerical data such as finance or climatology. This bot is making 10 day forecasts on financial assets whose data is sourced from Yahoo Finance because the model is using Yahoo Finance's API. I created a website for sharing my research project with everyone (hamdiai.com). In the website I am also showing the accuracy rates for every stock. These accuracy scores are determined by some flexible rules mostly focusing on the direction without ignoring the amount change. These rules are mostly created by some situations that happened on the trial of the bot on Investopedia Simulator. The test data includes days between 01.01.2025 and today. As it could be seen in the website the accuracy rates are between 70% to 80% for some well-known stocks. This accuracy rate for a 1.5 year timeframe is evidence of the power of AI on the Stock Market. Still when we think about this project being made by a 17 year old student with limited resources, it raises a haunting question: What can the huge banks or fund companies create with nearly infinite computers and software teams? On the other hand these accuracy rates are making people curious about "How does financial AI bots work? What is the analysis mechanism behind these forecasts?"

Unfortunately we do not know the calculations behind the system because we are only providing numerical data to the model and naming them so it is finding and learning the sequences and the repetitive data of the named number lists. The process is impossible for humans to do or understand because the LSTM models are designed for doing a lot of calculations in time series. At this point we can define the financial AI bots as "Black Box Models". A black box model is a system where the user just gives the input data and the system processes a final output but through this operation the user does not know how the system came up with that result. The advantage of these models is that they can process data pretty fast.[1] However these models create a controversial ethical question: In an automated market that has its calculations hidden inside a black box to what extent can we still consider the human factor? The black box model system not only does not involve any human based intuition also does not allow humans to recognize the mathematics behind. The human factor being excluded this much from finance is capitulating the control. In the project that I have made I could see that the outputs are accurate but it is impossible to see the calculations because its aim is to find the sequences between different macro economical data in a 5 year time period, so even if we could see the calculations by simple math equations it would probably be too complex for humans to understand. In this context my AI bot helped me to understand the notion of black box, by letting me have a first person experience to create and test a model.

As it was mentioned, I created a pretty accurate AI bot with limited resources and knowledge. It means that if someone or a group who has better computers, more education and experience than me create a model it would probably be a lot more accurate than mine. So, when banks train AI bots they can

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[1] Kenton, Will. "Understanding Black Box Models: Definition, Finance Use, and Examples." *Investopedia*, Investopedia, www.investopedia.com/terms/b/blackbox.asp. Accessed 16 Mar. 2026.

make lucrative investments with less risk. As a result of this the power holders will be earning more money and power as the time passes by. When someone makes money, some other person must lose money so that the system can stay balanced. If rich people create better AI models that means they will make more money which will cause normal people to lose money. So, the gap between rich and poor will widen. The problem about this scenario is that there is nothing middle and lower class people can do to hinder this situation. The effect of AI being capable of making highly accurate forecasts is creating inequality of opportunities. People who think that everyone has access to artificial intelligence would have fallen into the trap of democracy illusion. There will likely be more powerful models working in the background. Working and studying on my own project lead me to understand the potential of AI. Banks and companies would be utilizing this potential and will have a better chance to win against the normal people who are trying to search for companies and analyze them because based on my experience humans will not be able to compete with artificial intelligence in finance. At the end human labour is losing value caused by this fraudulent system.

To sum up, my experience showed me that finance is becoming the battlefield of artificial intelligence algorithms. The human factor is slowly being removed from finance. Artificial intelligence is gradually hiding finance into a black box. The gap between the middle class and the wealthy is increasing by the power of artificial intelligence. In a world where the human factor is losing its control and effectiveness, the notion of social justice fades away. Financial markets should not be a place where wealthy people make money out of middle class people with an unfair advantage such as more accurate AI bots. Markets should be fair for everyone with the same opportunities. If we do not optimize these artificial intelligence models with ethical principles, then probably we will completely lose control and reach the point of no return.